

AareML

Predicting River Water Quality in Swiss Catchments

CAS Advanced Machine Learning — Final Project

The Problem

- Dissolved oxygen (DO) is the primary stress indicator for river ecosystems — drops below 6 mg/L cause fish stress
- Swiss rivers face increasing thermal stress from climate change
- Current monitoring is reactive — no multi-day predictive capability

14-day

forecast horizon

early warning window

Data & Experimental Setup

Dataset

CAMELS-CH-Chem

(Nascimento et al. 2025)

- 17 Swiss gauges
- Daily records: 1981–2020
- 4 features: temp, pH, EC, DO

Experimental Setup

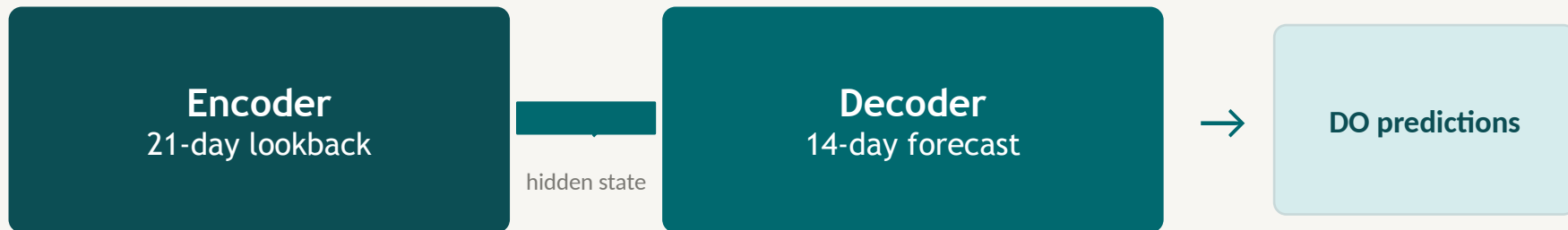
Split:

- Train 1981–2014
- Val 2015–2016
- Test 2017–2020

Model:

- Seq2Seq LSTM
- LOOKBACK = 21 days
- HORIZON = 14 days

Seq2Seq LSTM Architecture



Key Design Choices

- Teacher forcing with linear decay schedule
- Optuna hyperparameter search — 75 trials
- 3-seed ensemble for variance estimation

Optuna Search Space

- hidden_size: 64-256
- num_layers: 1-3
- dropout: 0.1-0.4
- lr: 1e-4 - 1e-2

Single-Site Results — Gauge 2473

Model	RMSE (mg/L)	KGE
Persistence	0.339	0.930
Climatology	0.332	0.853
Ridge Regression	0.303	0.908
VAR(7)	0.299	0.911
LSTM (Optuna) ★	0.303	0.945

Key Insight

LSTM matches VAR(7) on RMSE but leads on KGE

0.945 vs 0.911

Superior distributional fidelity

Zero-Shot Transfer to 11 Swiss Gauges

0.451 mg/L

mean zero-shot RMSE

0.568 mg/L

Ridge zero-shot

baseline comparison

0.390 mg/L

Per-gauge retrain

with fine-tuning

0.435 mg/L

EA-LSTM

with static attributes

Wilcoxon: $p = 0.037$, $n = 10$ (uncorrected)

Temperature Prediction

DO Temperature Forecast

Zero-shot

EA-LSTM

2.598°C → 1.360°C

47% improvement

EC Proxy Analysis

Temp + EC features

1.414°C RMSE

Temperature-only

1.451°C RMSE

2.6% gain from adding EC as proxy feature

Cross-Ecosystem & Cross-Continental Transfer

Lake Mendota (nb06)

2.962 mg/L

Zero-shot RMSE

NSE = -2.145 — FAILED

Swiss Lakes (nb10)

0.768 mg/L

Retrained RMSE

Zero-shot: 3.980 mg/L → retrained ✓

USGS US Rivers (nb08)

1.368 mg/L

Mean zero-shot RMSE

4.5× harder than Swiss rivers

Key Finding: Domain-specific retraining essential for lake transfer

Independent Benchmarks

NeuralHydrology EA-LSTM

Multi-gauge (11 sites):

- Mean RMSE: 0.512 mg/L
- NSE: 0.756

Gauge 2473:

- RMSE: 0.407 mg/L, NSE: 0.816

Supports feasibility of seq2seq approach

Cascaded Model (DO = linear(T) + residual)

Gauge 2473 comparison:

- Setup A: 0.872 mg/L
- Setup B: 0.893 mg/L
- **LSTM: 0.303 mg/L**

LSTM already captures Henry's Law implicitly — no cascading needed

Conclusions & Next Steps

Key Findings

- Seq2Seq LSTM: RMSE = 0.303 mg/L, KGE = 0.945 at gauge 2473 — comparable to VAR(7) on RMSE, superior on dynamics
- Zero-shot transfer to 11 gauges: 0.451 mg/L ($p = 0.037$ vs Ridge, uncorrected)
- EA-LSTM temperature: 47% improvement over zero-shot (1.360°C)
- Lake transfer fails without retraining — ecosystem boundaries matter

Next Steps

- Real-time AWEL data integration
- PFAS monitoring pipeline
- 7-day operational system (AquaSentinel AI)